Enhancing Hyperspectral Image Classification: Leveraging Unsupervised Information With Guided Group Contrastive Learning

Ben Li, Leyuan Fang, *Senior Member, IEEE*, Ning Chen, Jitong Kang, and Jun Yue

***Abstract*— Deep learning (DL) has demonstrated remarkable performance in the classification of hyperspectral images (HSIs) by leveraging its powerful ability to automatically learn deep spectral–spatial features over the years. Nevertheless, the limited supervisory signals along with a vast number of parameters in deep models still pose critical challenges when utilizing a restricted number of samples for training deep networks. To better handle this issue, this article proposes an end-to-end framework called guided group contrastive learning (GGCL) that adaptively integrates unsupervised information into a supervised contrastive learning framework. The proposed method employs a similarity-guided module that measures the spectral–spatial similarity of unsupervised samples based on supervised signals and effectively groups them. Then, the similarity signals of both supervised and unsupervised data are combined with contrastive learning to achieve intragroup feature aggregation and intergroup feature separation with guided group contrastive loss (GGCLoss). The pivotal characteristic of the proposed method lies in the end-to-end incorporation of unsupervised information with supervised signals for contrastive learning. Experiments on three public HSI datasets demonstrate that the proposed method can achieve better performance than existing state-of-the-art (SOTA) methods. For ease of reproducibility, the code of the proposed GGCL will be publicly available at https://github.com/fanerlight/GGCL_HSI.**

***Index Terms*— Contrastive learning, deep learning (DL), end-to-end framework, hyperspectral image (HSI) classification, spectral–spatial similarity.**

I. Introduction

HYPERSPECTRAL remote sensing technology represents a variegated and swiftly-evolving field, with the capacity to furnish valuable insights into the spectral characteristics of diverse objects and substances on the Earth's surface [1], [2]. Specifically, hyperspectral image (HSI) is comprised of both spatial and spectral reflectance data associated with earthly objects; each pixel corresponds to a spectral curve. Significantly, the spectral detection range of hyperspectral technology far exceeds that of human perception, thereby enhancing humanity's comprehension of the surface of our planet [3]. Harnessing both spatial texture and spectral properties, HSI has been successfully and extensively applied in a manifold of areas, encompassing but not confined to agriculture, ecology, geology, and land management. Preeminent among the objectives of HSI classification is the accurate classification of each pixel into a corresponding category relevant to the surrounding ground object [4], [5]. As a crucial component of HSI data processing, HSI classification fulfills an indispensable and irreplaceable role across the majority of hyperspectral technology applications [6], [7].

The high-dimensionality, redundancy, and noise in hyperspectral data present challenges for classification analysis [8], [9]. To address these challenges, numerous spectral dimensionality reduction techniques have been proposed, including linear discriminant analysis (LDA) [10], independent component analysis (ICA) [11], and singular value decomposition (SVD) [12]. Because of the nonlinear nature of HSI data, nonlinear transformation-based spectral feature extraction methods have been proposed in recent years, such as locality preserving projection (LPP) [13], [14] and nonnegative matrix factorization (NMF) [15]. The spatial heterogeneity and homogeneity present in hyperspectral imaging data make it challenging to effectively utilize spatial structure information and fully capitalize on the classification potential of spectral features alone. To fully exploit the spectral–spatial information, a series of joint spectral–spatial feature extraction methods have been proposed, including extended morphological profile (EMP) [16], directional morphological profile (DMP) [17], and extended attribute profile (EAP) [18].

Deep learning (DL), as a breakthrough in the field of machine learning, has been widely applied to various image analysis tasks [19], such as image segmentation [20], image classification [21], image fusion [22], image vectorization [23], and object detection [24], due to its rapid and significant development in recent years [25]. Concerning hyperspectral imaging classification, many researchers have proposed DL-based methods to enhance its accuracy. To fully utilize the

Manuscript received 26 September 2023; revised 30 November 2023; accepted 31 December 2023. Date of publication 8 January 2024; date of current version 19 January 2024. This work was supported in part by the National Natural Science Foundation of China under Grant U22B2014 and Grant 62101072, in part by the Science and Technology Plan Project Fund of Hunan Province under Grant 2022RSC3064, and in part by the Peking University's Undergraduate Teaching Reform Project under Grant JG2023019. *(Ben Li and Leyuan Fang contributed equally to this work.) (Corresponding authors: Ning Chen; Jun Yue.)*

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Digital Object Identifier 10.1109/TGRS.2024.3350700

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spectral–spatial features of HSIs, several joint spectral–spatial HSI feature extraction methods based on deep neural networks have been proposed, including stacked autoencoders [26], spatial pyramid pooling [27], fully convolutional network [28], transformers [29], [30], and graph convolutional network [31], [32]. These methods all leverage the power of DL to effectively extract spectral–spatial features from HSIs, and have achieved remarkable performance in HSI classification tasks [33].

However, acquiring sufficient labeled data for DL in practice is often a challenging task, prompting researchers to explore methods of extracting more discriminative spectral–spatial features from limited labeled samples to improve classification accuracy [19]. In addition, researchers have explored incorporating unlabeled samples as supplementary information to labeled samples within the model. It has been discovered that utilizing extra unlabeled samples through semisupervised learning algorithms can effectively enhance classification accuracy. Semisupervised learning algorithms encompass multiple types, such as self-training [34], co-training models [35], [36], generative models [37], [38], and graph-based methods [32], [39]. Furthermore, contrastive learning, as an unsupervised learning method, is becoming an important component in the field of semisupervised learning.

Contrastive learning [40] is a machine learning technique that is based on measuring sample similarity and identifying relationships between them [41]. Unlike traditional supervised learning, contrastive learning does not require standard class labels or a large amount of annotated data [42], [43]. It exhibits strong generalization ability and robustness and can improve model performance when supervised learning data is insufficient [44]. Several HSI classification methods that utilize contrastive learning for pretraining and fine-tuning have been proposed [45], [46], [47], [48]. These approaches place a reliance on well-designed data augmentation techniques. However, it is acknowledged that such methods may lead to a situation where samples of distinct instances belonging to the same class may be allocated to a pair of negative samples [44]. Furthermore, while several supervised contrastive learning-based approaches for HSI classification have been proposed recently [49], leveraging unsupervised information remains a challenge. Specifically, it remains difficult to use supervised information to generate positive and negative sample pairs within unsupervised data.

Based on the above analysis, in order to simultaneously utilize the precise supervised signals and adaptively integrate unsupervised information, this article proposes guided group contrastive learning (GGCL), which can guide unsupervised data based on supervised signals to generate positive and negative sample groups, fully utilizing the advantages of the accuracy of supervised information and the strong generalization ability of a large amount of unsupervised information. Our method can perform end-to-end training using both supervised and unsupervised samples, measure the similarity of unsupervised information, and group it through the similarity-guided module. Furthermore, by introducing the guided group contrastive loss (GGCLoss), the output of the similarity-guided module is combined with the contrastive loss to achieve intragroup feature aggregation and intergroup feature separation, effectively alleviating the problem caused by mixing false negative samples and ultimately improving the performance of the HSI classification model.

The proposed GGCL is an end-to-end supervised contrastive learning framework. Compared to most two-stage contrastive learning methods, our method integrates non-end-to-end methods into a single model, reducing intermediate steps, complexity, and inconsistency, while also integrating supervised information properly. Compared to existing end-to-end methods, we are able to make full use of unsupervised information. The main contributions of our proposed method are outlined as follows.

1) This article introduces a novel contrastive learning framework aimed at achieving end-to-end training for HSI classification by simultaneously utilizing supervised and unsupervised information. By integrating unsupervised information, the proposed framework alleviates overfitting caused by limited samples.
2) This article proposes a similarity-guided module that leverages supervised information to generate spectral–spatial similarity for guiding and grouping unlabeled samples. By combining spectral–spatial similarity with GGCLoss, the module achieves both intragroup feature aggregation and intergroup feature separation, thereby enhancing the exploitation of unsupervised information.
3) The proposed method is an end-to-end framework that can adaptively incorporate unsupervised information into contrastive learning. With the progress of training, the spectral–spatial similarity information of unsupervised samples gradually converges, ultimately improving the classification performance. Empirical evaluation shows that this method achieves high accuracy on multiple datasets.

The remaining sections of this article are organized as follows. In Section II, the related work of this study's methodology is introduced, including HSI classification and contrastive learning. In Section III, the proposed method is described in detail, including GGCL, GGCLoss, and spectral–spatial transformer encoder. In Section IV, the experimental datasets and training details are described, and the HSI classification results are presented and discussed, along with ablation studies being implemented and analyzed. Finally, Section V provides a conclusion of this article, and offers comments and suggestions for possible future research directions.

II. Related Work

As mentioned, DL has exhibited remarkable performance in the classification of HSIs due to its powerful capability of automatically learning deep spectral–spatial features. However, further research is needed to explore DL-based HSI classification methods. One major challenge is the requirement of a large number of labeled samples for effectively training deep models. Semisupervised learning algorithms provide a feasible solution. Recently, algorithms in the field of semisupervised learning have evolved into various types. Self-training is a typical semisupervised learning algorithm [34], where a classifier is first trained on a small amount of labeled

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data, and then used to classify unlabeled data. Subsequently, the most confident unlabeled samples are selected, and their predicted labels are added to the training set. Co-training models [35], [36] describe unlabeled samples from different perspectives to effectively utilize unsupervised information. Generative models [37], [38] train corresponding generators to construct more virtual samples to assist in training the classification model. Graph-based methods [32], [39] first represent each sample as a node in an affinity graph, from which the label information of unlabeled samples can be inferred based on the structure of the constructed graph. Currently, contrastive learning, as an unsupervised learning method, is becoming an important component in the field of semisupervised learning. Our article will focus on discussing contrastive learning algorithms.

The basic idea of contrastive learning is to make the distance between similar samples as close as possible while pulling apart the distance between different samples of categories. Generally, contrastive learning uses several loss functions to characterize the distance between similar and dissimilar samples. Common contrastive loss functions in the field of computer vision include normalized temperature-scaled cross entropy (NT-Xent) loss [40] and noise contrastive estimation (NCE) [50]. An important feature of contrastive learning is that it can learn without requiring labeled data. This feature enables contrastive learning to cope with some special problems, such as limited labeled data or high annotation costs, thereby inspiring and tapping the potential of data. It is crucial to select suitable positive and negative samples for the effectiveness of contrastive learning, so extensive research has been conducted on the methods of generating positive and negative samples. The main method is using the instance discrimination proxy task to generate positive and negative samples [51], [52], [53]. Meanwhile, data augmentation plays an important role in the process of generating positive and negative samples [40], [54], [55]. There are also some works that generalize some methods of contrastive learning as a dictionary lookup problem [56], [57]. In recent years, methods without negative samples have also been proposed [58], [59]. After the advent of the vision transformer, many contrastive learning tasks have used the vision transformer as a backbone for contrastive learning because it greatly improves the performance of the encoder [60], [61].

In the field of HSI classification, current research on contrastive learning can be broadly classified into two categories: two-stage methods based on pretraining and fine-tuning [45], [46], [47], [48] and end-to-end methods. Both of these approaches have limitations as follows.

1) The pretraining and fine-tuning methods cannot achieve end-to-end information extraction, and their negative sample generation method easily leads to the separation of similar samples [44].
2) End-to-end supervised contrastive learning methods currently face challenges in utilizing unsupervised information.

Therefore, this article proposes a contrastive learning scheme that can simultaneously utilize supervised and unsupervised information in an end-to-end manner to address the above issues. This scheme can use supervised information to generate unsupervised samples as contrastive learning supervision signals in an end-to-end manner to fully utilize the advantages of supervised and unsupervised learning.

III. Methodology

In this section, we will first introduce the overall framework of our proposed model. Then, we will explain the concept and details of GGCL. Finally, we will provide a detailed introduction to the spectral–spatial transformer encoder, which is responsible for extracting spectral–spatial features.

A. Overview

The overall framework of our proposed method is illustrated in Fig. 1. Given an HSI $\mathcal{H} \in \mathbb{R}^{H\times W\times B}$, where H and W denote the height and width of $\mathcal{H}$, respectively. B represents the number of spectral channels. Assuming the entire HSI dataset consists of N labeled pixels $X_N = \{x_1, x_1, \ldots, x_N\} \in R^{1\times 1\times B}$ and M unlabeled pixels $X_M = \{x_1, x_2, \ldots, x_M\} \in R^{1\times 1\times B}$. Each labeled pixel has a class label expressed in one-hot form, $Y_N = \{y_1, y_2, \ldots, y_N\} \in R^c$, where c represents the number of land cover classes. Neighboring cubes centered at pixels in X_N and X_M form a new group of dataset $Z_N = \{z_1, z_2, \ldots, z_N\} \in R^{w\times w\times B}$ and $Z_M = \{z_1, z_2, \ldots, z_M\} \in R^{w\times w\times B}$. To effectively leverage the spectral and spatial information available in HSIs, the proposed networks adopt cubes as input.

Our model combines labeled and unlabeled data in a certain proportion as model input, $Z_I^0 = \{z_i^n, z_j^m | z_i \in Z_N, z_j \in Z_M\}$, where m and n represent the number of unlabeled and labeled data instances mixed in, respectively. The mixing ratio m/n varies during the training process and is controlled by (1), where mix_fun$(\cdot)$ can be a piecewise function or linear function. In this study, a piecewise function was chosen as (2). To effectively leverage contrastive learning for better model performance, we apply data augmentation to the model input Z_I^0, resulting in two augmented inputs, Z_I^1 and Z_I^2. The augmentation is performed by applying odd-even masks in the spectral dimension, as shown in Fig. 2. For a specific patch $z^0 \in Z_I^0 \in R^{w\times w\times B}$, where B denotes the total number of spectral channels. We denote each channel as c_b, and $z^0 = \{c_b | b = 0, 1, .., B\}$. We extract the channels with odd spectral indices to form $z^1 = \{c_b | b = 1, 3, 5, ..\}$, and similarly, we extract the channels with even spectral indices to form $z^2 = \{c_b | b = 0, 2, 4, ..\}$. In this way, we transform z^0 into two sets of input data with nonoverlapping channels

$$m/n = \text{mix_fun}(\text{epoch}) \tag{1}$$

$$m/n = \begin{cases} 0 & \text{epoch} <= 100 \\ 1 & \text{epoch} > 100. \end{cases} \tag{2}$$

On one hand, Z_I^0 undergoes spectral–spatial feature extraction utilizing the spectral–spatial transformer encoder. Subsequently, it undergoes further processing through the classification tower, resulting in the output $Z_O^0 \in R^{(m+n)\times c}$, encompassing both labeled and unlabeled data. As shown in (3), the model selectively filters the labeled data and

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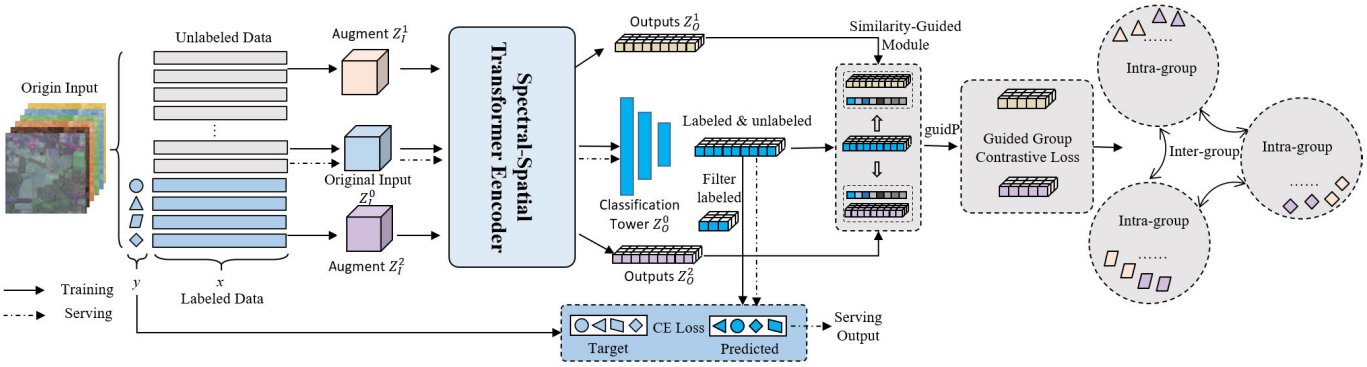


Fig. 1. Overview of the proposed GGCL.

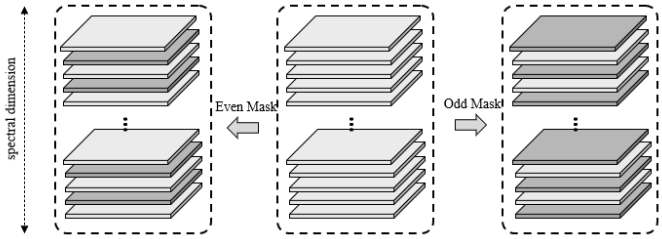


Fig. 2. Data augmentation: applying odd-even mask to the data in spectral dimension.

calculates the cross-entropy loss (CELoss) by comparing it with the actual label Y_N, where p_{ij} represents the probability values obtained by applying softmax to Z_O^0

$$\text{CEloss} = -\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{c}\left(y_{ij}\log\left(p_{ij}\right)\right). \tag{3}$$

On the other hand, the augmented inputs Z_I^1 and Z_I^2 are processed through the spectral–spatial transformer encoder structure with shared parameters, resulting in the outputs Z_O^1 and Z_O^2, respectively. At this stage, Z_O^1 and Z_O^2 contain both labeled and unlabeled data. Next, Z_O^0, Z_O^1, and Z_O^2 are fed into the similarity-guided module for transformation. This module utilizes the classification output information from Z_O^0 to guide Z_O^1 and Z_O^2, obtaining corresponding guided weights. Finally, the GGCLoss is computed through a contrastive loss. The GGCLoss will be discussed in detail in Section III-B. As shown in (4), the overall loss of the model is obtained by combining the classification loss and the GGCLoss with a weighted sum. The weighting factor is denoted as λ. These two losses mutually complement each other, jointly improving the classification performance of the model

$$\text{loss} = \text{CEloss} + \lambda \cdot \text{GGCLoss}. \tag{4}$$

B. Guided Group Contrastive Learning

We propose the method of GGCL to fully leverage unsupervised information within the context of contrastive learning and achieve end-to-end training. GGCL consists of two key modules, namely, the similarity-guided module and the GGCLoss. The former measures the similarity of unlabeled data by incorporating information from the classification tower, effectively enhancing the information of unsupervised data. The latter combines the output of the similarity-guided module with contrastive loss, modeling the intergroup differences between labeled and unlabeled data through contrastive learning, ultimately improving the performance of the classification model.

1) Similarity-Guided Module: The core of the similarity-guided module is to provide gw $\in R^{(m+n)\times(m+n)}$ (guide weight) for computing GGCLoss. To calculate gw, it is necessary to compute guidP $\in R^{(m+n)\times c}$, as shown in (5), which is generated based on the labels for labeled data and softmax probabilities from the output of the classification tower for unlabeled data. We use guidP as input and map it to gw using the given guid_fun as (6). There are various choices for guid_fun. Specifically, in this article, guid_fun is chosen as shown in (7). gw represents the importance of considering any two samples as belonging to the same class. The larger the value of gw, the greater the similarity between the two samples in terms of belonging to the same class

$$\text{guidP}_{ij} = \begin{cases} y_{ij} & y_i \in Y_N \\ \text{softmax}\left(Z_o^0\right)_{ij} & y_i \notin Y_N \end{cases} \tag{5}$$

$$\text{gw} = \text{guid_fun}(\text{guidP}) \tag{6}$$

$$\text{gw}_{ik} = \begin{cases} 1 & \underset{j}{\text{argmax}}\left(\text{guidP}_{ij}\right) = \underset{j}{\text{argmax}}\left(\text{guidP}_{kj}\right) \\ 0 & \underset{j}{\text{argmax}}\left(\text{guidP}_{ij}\right) \neq \underset{j}{\text{argmax}}\left(\text{guidP}_{kj}\right). \end{cases} \tag{7}$$

2) Guided Group Contrastive Loss: We propose GGCLoss to combine the output of gw with contrastive loss, utilizing contrastive learning to model the intergroup differences between labeled and unlabeled data, ultimately improving the performance of the classification model. First, we compute the cosine similarity s_{ik} by (8) for any pair $< \text{vec}_i, \text{vec}_k >$ from Z_O^1 and Z_O^2. Then, we calculate GGCLoss by incorporating the gw obtained from the similarity-guided module and the pair similarity S into (9), where τ represents the temperature parameter. The main difference between GGCLoss and traditional contrastive loss lies in utilizing gw to group unsupervised information while considering the importance of different pair similarities. The pseudocode of training the

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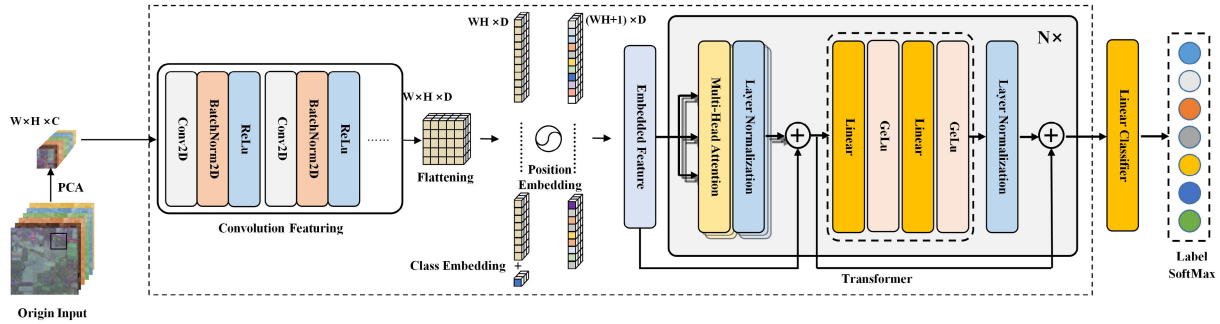


Fig. 3. Training and serving framework of our guided contrastive model.

GGCL model is shown in Algorithm 1

$$s_{i,k} = \frac{\text{vec}_i \cdot \text{vec}_k}{\|\text{vec}_i\| \|\text{vec}_k\|} \tag{8}$$

$$\text{GGCLoss} = -\sum_i \left(\sum_k \left(\text{gw}_{ik} \cdot \log \frac{\exp(s_{ik}/\tau)}{\sum_k \exp(s_{ik}/\tau)} \right) / \sum_k \text{gw}_{ik} \right). \tag{9}$$

C. Spectral–Spatial Transformer Encoder

Fig. 3 illustrates the basic structure of the spectral–spatial transformer encoder. As mentioned in Section III-A, the spectral–spatial transformer encoder takes Z_I^0, Z_I^1, and Z_I^2 as inputs. The input data first undergoes the Conv2D structure. The Conv2D structure is able to extract preliminary spatial information and compress the spectral dimension information. After the Conv2D structure, the model adds position encoding to incorporate positional information into the embedded features before entering the transformer structure. The transformer structure consists of multihead attention and multilayer perceptron (MLP), which are hierarchically stacked to obtain the output of the classification information. Finally, a linear classifier is used to produce the results of multiclassification. This structure combines the characteristics of both CNN and transformer. It initially utilizes the CNN structure for preliminary spatial and spectral information extraction and then employs the transformer structure for deeper fusion of spectral–spatial information, effectively improving the classification performance of the model.

Algorithm 1 GGCL Model

Require: Untrained GGCL model ***GM***, untrained classifying head ***CH***, dataset ***H***

Ensure: Trained Model ***GM*** and ***CH***

Extract limited labelled data X_N from ***H***: $X_N \subset \boldsymbol{H}$
Divide ***H*** into unlabelled data X_M: $X_M = \boldsymbol{H} - X_N$
for each Epoch **do**
  Sampling labelled data batch from X_N and unlabelled data batch by Eq. (1) $Z_I^0 = \{z_i^n, z_j^m | z_i \in Z_N, z_j \in Z_M\}$
  By the spectral odd-even mask, augment the unlabelled data: $Z_I^1 = odd(Z_I^0)$, $Z_I^2 = even(Z_I^0)$
  Generate the features $Z_O^1 = GM(Z_I^1)$, $Z_O^2 = GM(Z_I^2)$
  Input the labelled data to the model $Z_O^0 = GM(Z_I^0)$
  Calculate the CE loss by the labelled data by Eq. (3)
  Calculate the similarity of unlabelled data of $CH(Z_O^1)$ and $CH(Z_O^2)$ by Eq. (5) Eq. (7) and Eq. (6)
  Compute the cosine similarity for any pair from Z_O^1 and Z_O^2 by Eq. (8)
  Calculate the GGC Loss by Eq. (9)
  Calculate the total loss by Eq. (4)
  Update ***GM*** and ***CH*** by the total loss
end for
return ***GM*** and ***CH***

IV. Experiments Results

A. HSI Dataset

1) Indian Pines: The Indian Pines dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the Indian Pines agricultural site in Indiana, USA. This dataset comprises 145 × 145 pixels in size, with 224 spectral bands ranging from 400 to 2500 nm. Because the 104th to 108th bands and 150th to 163rd bands cannot be reflected by water, these bands are removed and the corrected bands are 200. The spatial resolution of the dataset is about 20 m. Indian Pines dataset is composed of various types of land covers including agricultural fields, forests, urban areas, and water bodies. These different land cover types make it an ideal dataset for research in HSI classification, spectral unmixing, and feature extraction. The availability of this dataset has contributed to the advancement of the research in hyperspectral imaging and remote sensing. The false color image and the classification ground-truth are shown in Fig. 4.

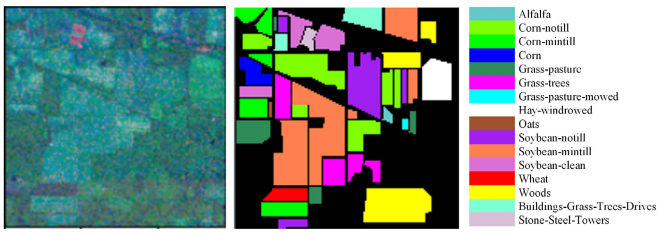


Fig. 4. False color map and the ground-truth land-use of dataset Indian Pines.

2) Pavia University: The Pavia University dataset is a high-resolution dataset of a university area in Pavia, Italy, captured by an airborne hyperspectral sensor. The dataset comprises 610 × 340 pixels in size, with 115 spectral bands ranging from 430 to 860 nm. In order to eliminate the affect

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TABLE I
TRAINING AND TEST SAMPLE NUMBERS IN THE INDIAN PINES DATASET, THE PAVIA UNIVERSITY DATASET, AND THE SALINAS DATASET

No.	Indian Pines			Pavia University			Salinas		
	Class	Train.	Test.	Class	Train.	Test.	Class	Train.	Test.
1	Alfalfa	10	36	Asphalt	10	6621	BrocoliGreenWeeds1	10	1999
2	CornMotill	10	1418	Meadows	10	18639	BrocoliGreenWeeds2	10	3716
3	CornMintill	10	820	Gravel	10	2089	Fallow	10	1966
4	Corn	10	227	Trees	10	3054	FallowRoughPlow	10	1384
5	GrassPasture	10	473	PaintedMetalSheets	10	1335	FallowSmooth	10	2668
6	GrassTrees	10	720	BareSoil	10	5019	Stubble	10	3949
7	GrassPastureMowed	10	18	Bitumen	10	1320	Celery	10	3569
8	HayWindrowed	10	468	SelfBlockingBricks	10	3672	GrapesUntrained	10	11261
9	Oats	10	10	Shadows	10	937	SoilVinyardDevelop	10	6193
10	SoybeanNotill	10	962				ComSenescedGreenWeeds	10	3268
11	SoybeanMintill	10	2445				LettuceRomaine4wk	10	1058
12	SoybeanClean	10	583				LettuceRomaine5wk	10	1917
13	Wheat	10	195				LettuceRomaine6wk	10	906
14	Woods	10	1255				LettuceRomaine7wk	10	1060
15	BuildingsGrassTreesDrives	10	376				VinyardUntrained	10	7258
16	StoneSteelTowers	10	83				VinyardVerticalTrellis	10	1797
	Total	160	10089	Total	90	42686	Total	160	53969

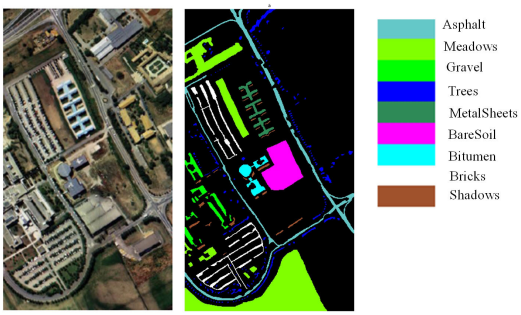


Fig. 5. False color map and the ground-truth land-use of dataset Pavia University.

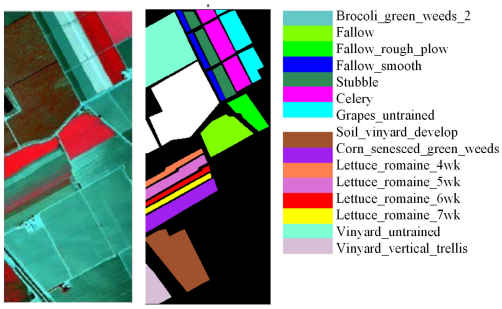


Fig. 6. False color map and the ground-truth land-use of dataset Salinas.

from noise, 12 bands are removed and the remained 103 bands are available. The spatial resolution of the dataset is 1.3 m. Pavia University dataset includes nine land covers such as asphalt, meadows, trees, and bare soil. It has been widely used in classification, detection, and recognition applications. The availability of this dataset has contributed to the advancement of the research in hyperspectral imaging and image analysis. The false color image and the classification ground-truth are shown in Fig. 5.

3) Salinas: The Salinas dataset was captured by an airborne hyperspectral sensor over an agricultural area in Salinas Valley, CA, USA. The dataset consists of 512×217 pixels in size, with 224 spectral bands ranging from 400 to 2500 nm. The spatial resolution of the dataset is about 3.7 m. Salinas dataset includes 16 different crop types such as lettuce, broccoli, onion, and cauliflower. It has been widely used in classification, clustering, and feature extraction applications. The availability of this dataset has contributed to the advancement of research in hyperspectral imaging and agricultural monitoring. The false color image and the classification ground-truth are shown in Fig. 6.

To satisfy the criteria of the small sample model, we impose strict limits on the number of training and testing samples in the dataset. This entails ensuring that each category's number of samples is equal in every dataset and dividing the dataset into distinct subsets based on the training sample count for each category. These counts include 10, 20, 30, 40, 50, 60, 70, and 80 samples. The residual samples are set aside for testing. For instance, Table I displays the breakdown of the dataset into ten equal training samples for each category.

B. Experimental Settings

1) Hyperparameters: The hyper-parameters of the proposed model include the patch size of inputs, the number of self-attention heads, and the number of transformer blocks. First, the patch size of input images is an important factor affecting the spatial feature extracting. Compared with smaller patches, bigger patches cover more neighbor information but increase the difficulty of featuring. The patch size has been set to 7, 9, 11, 13, and 15. After conducting experiments, we believed that the optimal choice of patch size for all datasets is 13×13, shown in Fig. 7. Second, the number of self-attention heads is related to the dimension of the data property. We have set it as 8, 16, 24, and 32, and we found that the optimal value of attention heads varied with datasets. The optimal values of heads are 24, 8, and 16 for the IP, PU, and SA datasets, respectively. We believed that this was related to the number of spectral bands. Third, the number of transformer blocks is related to the depth of extracted data information. The optimal values of block number are 2, 1, and 3 for the IP, PU, and

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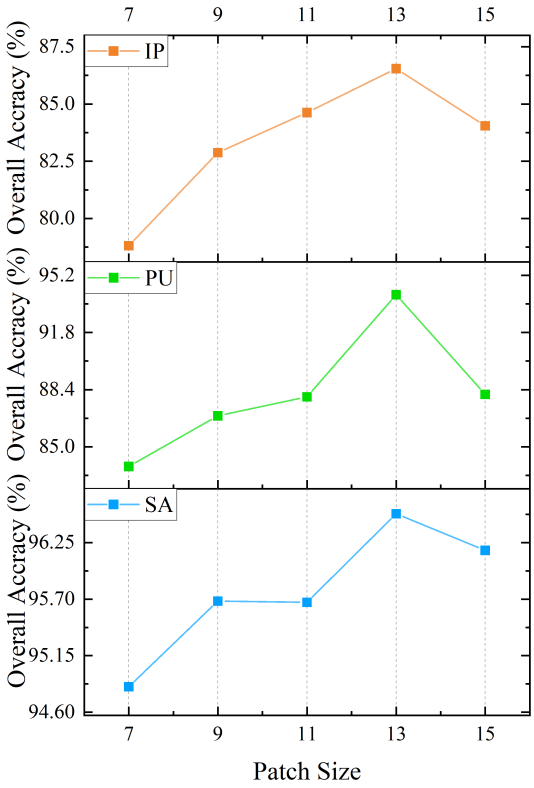


Fig. 7. Effect of patch sizes on the OA metrics.

SA datasets, respectively. The effect of transformer blocks and heads on the overall classification accuracy is shown in Fig. 8. Hyperparameters used in the training process, such as the batch size, the learning rate, and the number of training epochs, play an important role in the model's fitting quality. Generally, a bigger batch size tends to have more significant requirements for the hardware platform and higher difficulty for the model's learning process. In this article, we consider both the available platform and the learning difficulty, setting the batch size as 64, the learning rate as 0.001, and the number of training epochs as 200.

Another crucial hyper-parameter is the weighting factor λ in (4). In this equation, λ plays a role in adjusting the weight between the guiding CELoss and the group contrastive loss (e.g., GGC loss). The optimal value of λ is 0.1, and its tuning is a tradeoff process: when λ value is small, the contribution of a large amount of unsupervised information to the model is small, so the significance of contrastive learning is impaired; When the λ value is large, the guiding CELoss may be ignored, leading to the confusion of grouping by the contrastive learning. Therefore, to obtain the optimal λ value needs multiple experiments. We set λ as 0.01, 0.02, 0.05, 0.1, 0.2, and 0.5 for comparison, and recorded the OA, AA, and κ as evaluation metrics. The results are shown in Fig. 9.

As can be seen from Fig. 9, with the λ increases from 0.01 to 0.5, the OA, AA, and κ on all three datasets have met their peaks. When $\lambda = 0.1$, the OA and κ reach their peaks, and the AAs are also close to their peaks. Therefore, we set $\lambda = 0.1$ as the hyper-parameter of the model training.

Based on the optimal selection of hyperparameters, the number of parameters in the GGCL model can be calculated.

TABLE II
NUMBERS OF PARAMETERS IN THE PROPOSED GGCL MODEL

Dataset	CNN Section	Transformer Section	MLP Section	Total
IP	61696	401408	1024	464128
PU	61696	139264	576	201536
SA	61696	270336	1024	873728

The model backbone proposed in this article consists of a CNN section, a transformer section, and an MLP section. Among them, the parameter quantity of the CNN part is related to the convolutional kernel size k and the number of input channels *inChannel*, output channel *outChannel*. In this article, the CNN section includes two concatenated convolutional kernels with sizes 3 and 1, respectively. The changes in channel numbers are 100, 64, and 64, so the parameter quantity of the CNN section is $k \times k \times$ inChannel $\times$ outChannel + outChannel $\times$ outChannel. The parameter quantity of transformer blocks is related to the number of the transformer section, depth, number of attention head heads, input dimension dim, internal dimension innerDim, and linear layer dimension hiddenDim. The parameter quantity formula for this section is: depth $\times$ (heads $\times$ (dim $\times$ innerDim $\times$ 3 + innerDim $\times$ dim) + dim $\times$ hiddenDim + hiddenDim $\times$ dim). The parameters used in the experiment dim = 64, innerDim = 64, and hiddenDim = 64. The number of blocks and attention heads used in different datasets varies. In IP, depth = 2, heads = 24; in PU, depth = 1, heads = 8; in SA, depth = 3 and heads = 16. The MLP section is relatively simple, consisting of a single-layer linear layer, with parameter quantity and input dimension dim and classification quantity numClasses. Regarding classes, the parameter quantity formula is: dim $\times$ numClasses, where dim is 64 and numClasses differs for different datasets (IP16, PU9, and SA16). Based on these formulas, the parameter quantities of the GGCL model can be calculated, as shown in Table II.

2) Comparison With Other Models: To compare the performance of our contrastive learning model, we selected another eight representative methods of HSI classification. In order to be fair, all methods took the same training samples and testing samples.

1) The 2D-CNN are traditional basic DL methods, which are the base of all CNN-based models (e.g., our backbone structure). As a result, we kept the kernel sizes the same between the CNN methods and our backbone model. Specifically, our 2D-CNN architecture consists of three 2D-CNN blocks. Each block is comprised of a 3×3 2D-CNN structure, a BN layer, and a Rectified Linear Unit (ReLU) activation function. Finally, we utilize an MLP structure for classification.
2) The spectralformer [29] is a representative and popular application of the transformer model in HSI classification. Our model is also transformer-based, so it is meaningful to regard the spectral former as a baseline.
3) The spectral–spatial residual network (SSRN) [62] combines the CNN model and the residual network framework, and has achieved high accuracy in wide-used public datasets. The prominent advantage of SSRN

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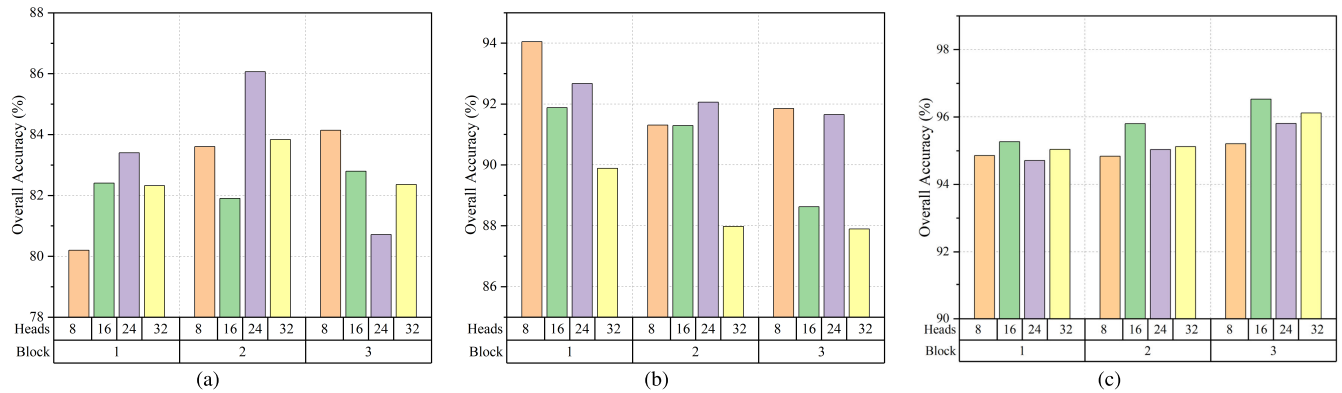


Fig. 8. Effect of transformer blocks and heads on the OA. (a) On the IP dataset. (b) On the PU dataset. (c) On the SA dataset.

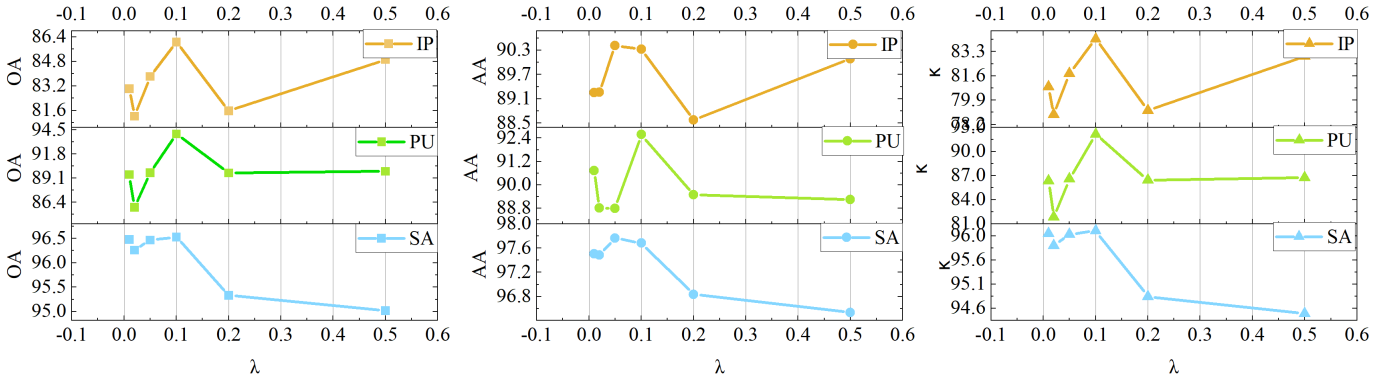


Fig. 9. Tuning of weighting factor λ.

is that it has alleviated the decreasing-accuracy phenomenon.

4) The spectral–spatial feature tokenization transformer (SSFTT) [30] utilizes the ability of spatial featuring and semantic featuring from the CNN model and the transformer encoder, respectively. And its Gaussian-weighted feature tokenizer helps generate high-level semantic tokens. It is one of the state-of-the-art (SOTA) HSI-supervised classification methods that are based on CNN and transformer.
5) The spectral–spatial graph reasoning network (SSGRN) [32] is an outstanding model on the HSI classification using the graph convolutions. It is selected to be compared with our model due to its SOTA effect among graph convolution network (GCN)-based methods.
6) The self-supervised contrastive efficient asymmetric dilated network (SC-EADNet) [47] claims that it makes full use of pixel-wise contextual information using a self-supervised multiscale feature learning model, and obtains competitive classification performance at fast computational speed using the asymmetric dilated convolution blocks in the contrastive learning framework.
7) The deep multiview learning and support vector machine (DMVL + SVM) [63] is a classical two-step contrastive learning model of HSI classification. In the first step, plenty of unlabeled data is used to form semantic features by contrastive learning. While on the second step, the SVM performs as the classifier on the off-the-shelf features.
8) The spectral–spatial Siamese network (S3Net) [49] is a coherent end-to-end model on few-shot HSI classification, using a weighted contrastive loss and adaptive CELoss. It utilizes a limited amount of labeled data but still has managed to alleviate the overfitting issue.

CNN and SSRN are classic CNN-based network structures, and SSRN in particular represents one of the SOTA CNN-based classification models. SF and SSFTT are both classic transformer-based network structures, and SSFTT in particular represents one of the SOTA transformer-based classification models. SSGRN is one of the latest published GCN-based and semisupervised algorithm network structures and has achieved SOTA performance. To compare with the currently most popular contrastive learning frameworks, we have selected DMVL, SC-EADNet, and S3net. The former two are classical two-stage contrastive learning algorithms, while the latter represents the performance level of the state-of-the-art end-to-end contrastive learning framework. The hyperparameters of the above models are set according to the optimal choices or the recommendations in the corresponding origin paper.

3) Evaluation Metrics: The overall accuracy (OA), average accuracy (AA), and Kappa coefficient are used to evaluate

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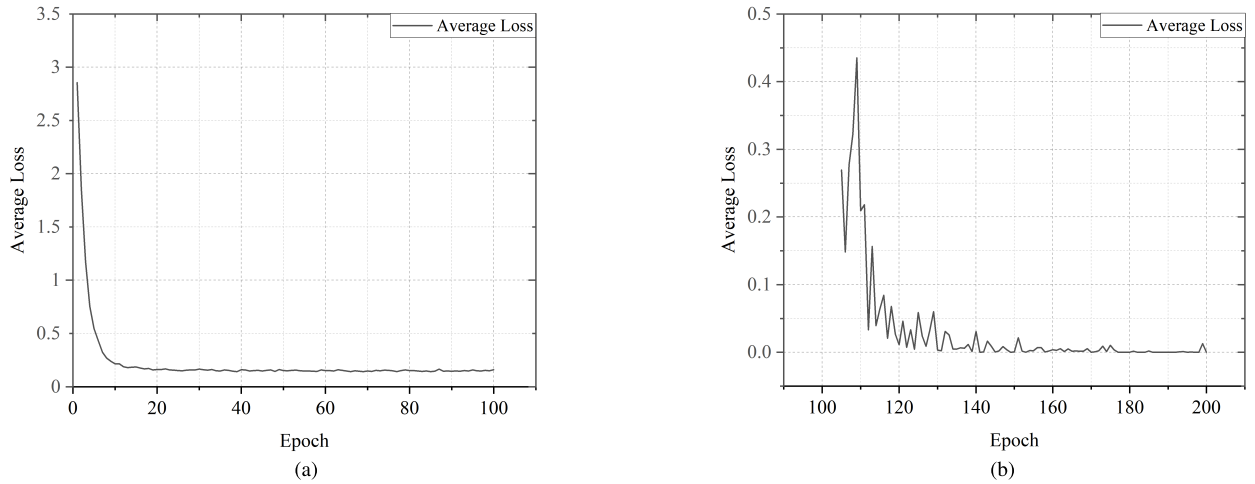


Fig. 10. Loss values during the training epochs. (a) Guiding loss. (b) Contrastive loss.

the performance of different models, and we report the best results after carrying out ten independent executions in every experiment.

4) Experimental Platform: All experiments are conducted on a platform with Nvidia GeForce RTX 3090 GPU and Intel[1] Xeon Gold 6234 CPU. The software platform is based on Python 3.8, Pytorch 1.9.1, and CUDA 11.7 toolkits.

C. Classification Results of Different Models

The convergence of the model can be seen from the decrease in the CELoss during the training process, as well as from the classification accuracy of the validation set. Because the CELoss directly reflects the classification results: the smaller the CELoss, the more accurate the classification. We recorded the average loss value for each epoch during the training process and validated every 10 epochs to obtain the corresponding OA. In order to better illustrate the changes in CELoss during the guidance and comparison processes, we divided the entire process of CELoss into the guidance process and the comparison process for display, as shown in Fig. 10.

It can be seen that the average loss of training during the guidance process decreases rapidly, and after 20 epochs, the loss tends to stabilize. The OA of the corresponding validation set also rapidly increased, but there was a slight decrease after 20 epochs. Therefore, the model converges during the guidance process. After the initial guidance is completed, the in-depth guidance process shows that although the loss fluctuates, it still maintains a downward trend; although the OA of the corresponding validation set fluctuates, it also shows an upward trend. Therefore, throughout the entire training process, the GGCL loss decreases, and the effect of the model is convergence.

The accuracy of 2D-CNN and SpectralFormer is unstable and relatively lower across all samples. To highlight comparison, we only evaluate the methods that have relatively high accuracies in previous performance comparison experiments, including SSRN, SSFTT, SSGRN, DMVL + SVM, S3Net, and ours model. The corresponding overall accuracies are shown in Fig. 11. It is evident that the model proposed in this study consistently demonstrates a high-level performance and presents a robust upward trend under the condition of a sample size between 10 and 80. Even though SSRN, SSFTT, SSGRN, DMVL + SVM, and S3Net exhibit a relatively high level of accuracy with an overall upward trend, they still fall below the model proposed. We provide a specific example of analyzing each class with only ten samples, as presented in Tables III–V.

From Table III, it can be seen that the proposed model in this article achieves the highest OA and κ value compared to another eight methods on the IP dataset. The highest values are 86.064% and 84.184, respectively, outperforming the second-ranking S3Net model by 0.367% and 0.387. It has the best performance for eight land-cover categories, tied with the SSFTT model. Specifically, for SoybeanMintill, which is the largest IP dataset, the proposed model demonstrates the highest classification accuracy compared to other models at 91.575%. The fact that the SoybeanMintill category has the smallest proportion of training samples, with only ten samples in each class, highlights the proposed model's superiority in small-sample classification. Based on Fig. 12, the proposed model displays better distinguishability in SoybeanMintill and CornMintill visually, surpassing the second-ranking S3Net model. In addition, the classification results of Woods, SoybeanMintill, and SoybeanClean are more comprehensive and closer to the ground truth in terms of spatial distribution than those of other models.

As shown in Table IV, the proposed model presents clear advantages on the PU dataset, with considerably higher OA, AA, and κ values than the other models, reaching 94.047%, 92.576%, and 92.115, respectively. In comparison to the second-ranking S3Net model, the proposed model achieved 3.877%, 0.777%, and 3.612 points higher in OA, AA, and κ values, respectively. This model exhibits the highest accuracy on the Meadows (18639) and Asphalt (6621) land-cover categories, reaching 97.1% and 98.138%, respectively, which is 3.7% and 1.36% higher than the second-ranking model. With only ten training samples per land-cover category, Meadows and Asphalt have the lowest proportion of training samples,

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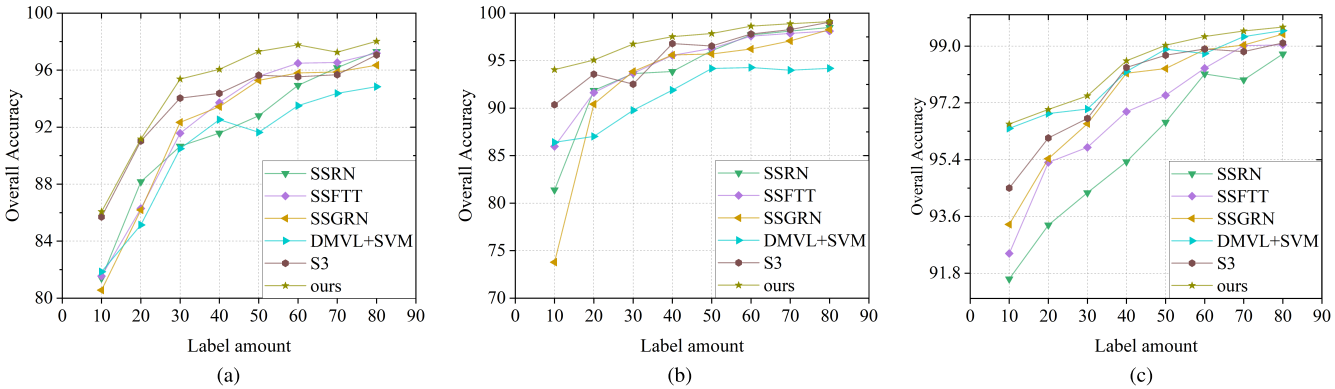


Fig. 11. OA using different amounts of labeled data per-class. (a) OAs on the Indian Pines dataset. (b) OAs on the Pavia University dataset. (c) OAs on the Salinas dataset.

TABLE III
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE INDIAN PINES DATASET (TEN LABELS PER-CLASS)

Class	2D-CNN	Spectral Former	SSRN	SSFTT	SSGRN	SC-EADNet	DMVL+SVM	S3Net	Ours
1	58.333	83.333	**100.000**	**100.000**	**100.000**	**100.000**	29.677	**100.000**	**100.000**
2	36.107	25.740	61.989	69.182	67.983	45.868	79.029	79.267	**82.299**
3	44.268	21.829	56.463	82.439	67.317	89.036	**96.519**	81.463	59.878
4	58.590	38.767	94.714	90.749	94.714	**100.000**	56.436	96.916	92.952
5	73.573	43.975	77.167	**95.349**	89.429	88.820	76.861	91.332	84.144
6	81.389	61.111	70.417	97.222	79.444	**99.589**	89.600	96.806	93.611
7	88.889	61.111	**100.000**	**100.000**	**100.000**	**100.000**	18.667	**100.000**	**100.000**
8	61.538	97.009	**100.000**	**100.000**	**100.000**	97.071	92.996	99.573	**100.000**
9	**100.000**	**100.000**	**100.000**	**100.000**	**100.000**	**100.000**	27.397	**100.000**	**100.000**
10	45.946	44.595	**92.827**	77.547	70.270	79.321	70.234	85.031	89.709
11	51.247	52.025	90.184	64.376	80.736	89.654	91.064	74.233	**91.575**
12	32.247	41.166	55.232	83.362	67.410	49.747	82.558	**90.909**	76.329
13	95.897	96.923	**100.000**	**100.000**	96.923	**100.000**	83.190	99.487	**100.000**
14	84.223	67.649	93.944	97.928	94.263	76.206	**99.548**	92.590	83.028
15	34.574	38.032	99.468	97.872	**99.734**	94.560	80.851	99.202	91.223
16	85.542	98.795	**100.000**	**100.000**	96.385	**100.000**	41.892	**100.000**	**100.000**
OA(%)	55.556	49.450	81.396	81.544	80.563	80.390	81.852	85.697	**86.064**
AA(%)	64.523	60.754	87.025	91.002	87.788	88.120	69.782	**92.926**	90.297
$\kappa \times 100$	50.007	43.252	78.984	79.246	77.997	77.790	79.488	83.797	**84.184**

demonstrating the proposed model's superior performance in small sample training tasks. According to Fig. 13, the proposed model shows a smoother spatial distribution and more accurate classification for Gravel and BareSoil types than the second-ranking S3Net model, with no small dots, which is closest to the ground truth.

According to Table V, the proposed model achieved higher OA, AA, and κ values than the other eight methods on the SA dataset, reaching 96.529%, 97.689%, and 96.141, respectively. Compared to the second-ranking DMVL + SVM model, it leads to 0.139%, 2.877%, and 0.035 higher. The slender advance on OA and κ could be due to the Salinas dataset's simplicity compared to the other two datasets, making it more suitable for a DMVL + SVM two-stage model. The proposed model achieved the highest accuracy in five land-cover categories, tying with the SSFTT and S3Net models for first place. Visual inspection, shown in Fig. 14, demonstrated the complete spatial distribution of the various land-cover categories in the SA dataset, characterized by clear boundary lines and simple polygon shapes. Our proposed model showed slight confusion in the Grapes_untrained and Vinyard_untrained land-cover categories. Conversely, the confusion level for the S3Net model was finer and more scattered in the same circumstances, indicating the good performance of our model in terms of spatial completeness and smoothness. Excluding Grapes_untrained and Vinyard_untrained, our model maintained a high level of consistency with the ground truth in the remaining 14 land-cover categories.

Based on the above results and analysis, it can be seen that our model has achieved SOTA classification performance. On the other hand, time efficiency and convenience are also key properties of a DL model. Therefore, all eight well-trained models were used to serve the three datasets, and their serving time spans were recorded. The result is shown in Table VI.

It can be seen that our model is much faster than many of the others, and is only slower than three models (i.e., 1D-CNN, 2D-CNN, and SSGRN). Compared with our model, the 1D-CNN and 2D-CNN lack of semantic embedding structure. Although they are faster, their classification performance

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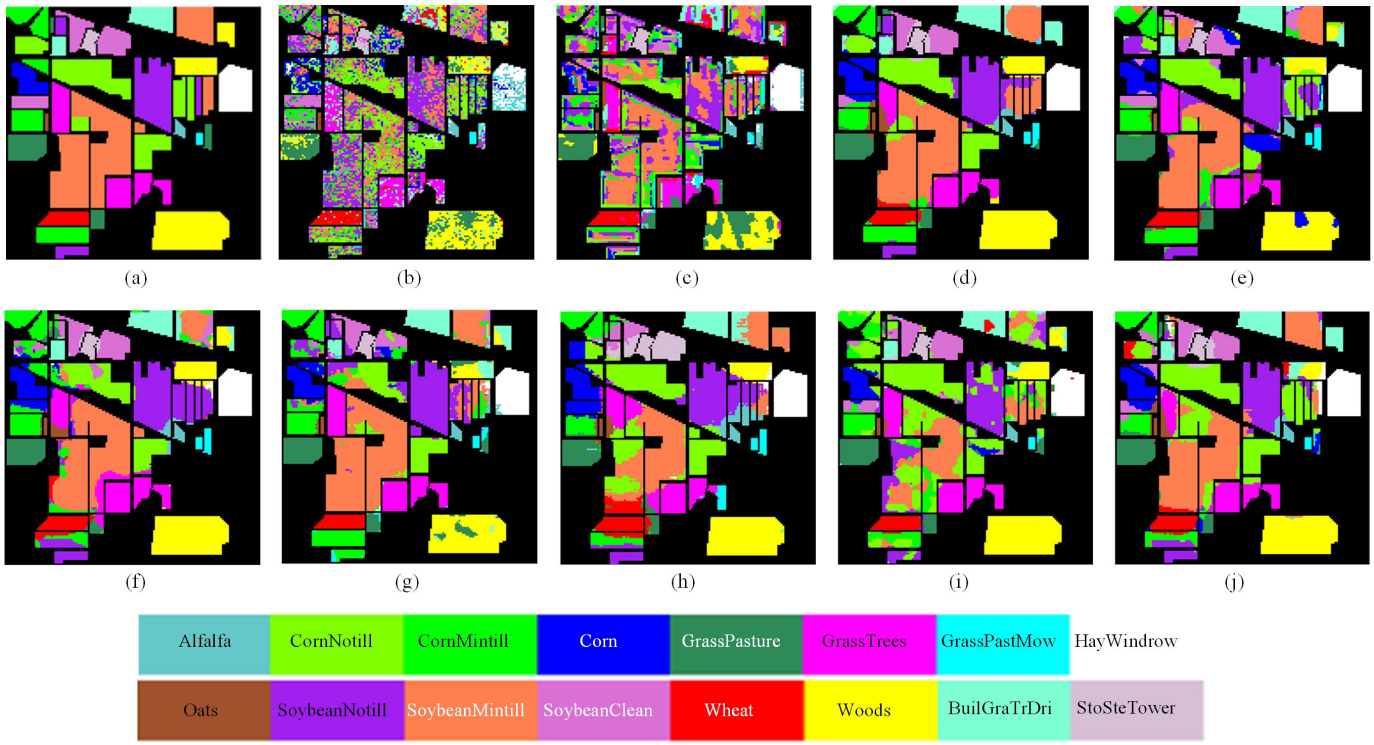


Fig. 12. Visual classification results of Indian Pines from compared methods [(a) Ground truth, (b) 2D-CNN, (c) spectral former, (d) SSRN, (e) SSFTT, (f) SSGRN, (g) SC-EADNet, (h) DMVL + SVM, (i) S3Net, and (j) ours].

TABLE IV
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE PAVIA UNIVERSITY DATASET (TEN LABELS PER-CLASS)

Class	2D-CNN	Spectral Former	SSRN	SSFTT	SSGRN	SC-EADNet	DMVL+SVM	S3Net	Ours
1	73.327	50.959	81.800	75.865	56.366	83.306	73.416	95.877	**97.100**
2	67.037	94.930	96.438	89.538	76.023	92.241	96.778	92.376	**98.138**
3	81.091	65.725	**93.585**	90.665	47.630	62.697	75.935	86.740	90.474
4	80.059	90.504	60.445	93.811	56.712	**98.727**	83.213	94.859	89.358
5	98.951	**100.000**	98.352	99.625	99.626	**100.000**	86.293	**100.000**	99.251
6	74.557	33.712	97.888	95.198	**98.127**	62.597	96.169	70.632	94.939
7	86.970	89.318	45.758	99.697	76.288	95.038	94.113	**100.000**	98.636
8	62.255	63.045	0.000	51.362	76.471	**95.139**	74.558	86.029	68.164
9	**100.000**	98.719	76.734	98.079	84.952	99.472	49.197	99.680	97.118
OA(%)	72.443	76.491	81.387	85.977	73.783	87.130	86.404	90.353	**94.047**
AA(%)	80.472	76.324	72.333	88.205	74.688	87.690	81.075	91.799	**92.576**
$\kappa \times 100$	65.165	68.426	75.081	81.849	66.862	82.920	82.295	87.236	**92.115**

is not good. In addition, compared with the DMVL + SVM and the S3Net which also belong to the supervised contrastive learning method, our model has better execution efficiency. The reason why our model is faster than the DMVL + SVM and the S3Net may be due to different reasons. Compared to the DMVL + SVM, our model has relatively simple structures so it requires less computation. Compared to the S3Net, our model avoids the process of data augmentation when serving the test dataset.

D. Model Analysis

1) Ablation Analysis: To further validate the superior performance of the proposed model's similarity-guided module and contrastive learning part in small-sample scenarios, we conducted ablation experiments. During these experiments, we maintained the same network framework, hyperparameters, and training set division. We only modified whether to include the crucial contrastive learning and guided intergroup module in the training process. Thus, we were able to illustrate the benefits and improvements brought by both modules through experimental results.

The ablation experiment was specifically designed as follows. First, we measured the efficacy of contrastive learning as a means of extracting information from unlabeled data. In Fig. 1, the data underwent Augment1 and Augment2 to form contrastive groups. The raw data, without augmentation, was directly filtered and underwent CELoss with labeled data. In Ablation Experiment 1, we excluded contrastive learning and only used CELoss between the raw and labeled data to train the network.

Second, we conducted a comparative analysis to evaluate the benefits provided by the similarity-guided module in contrast

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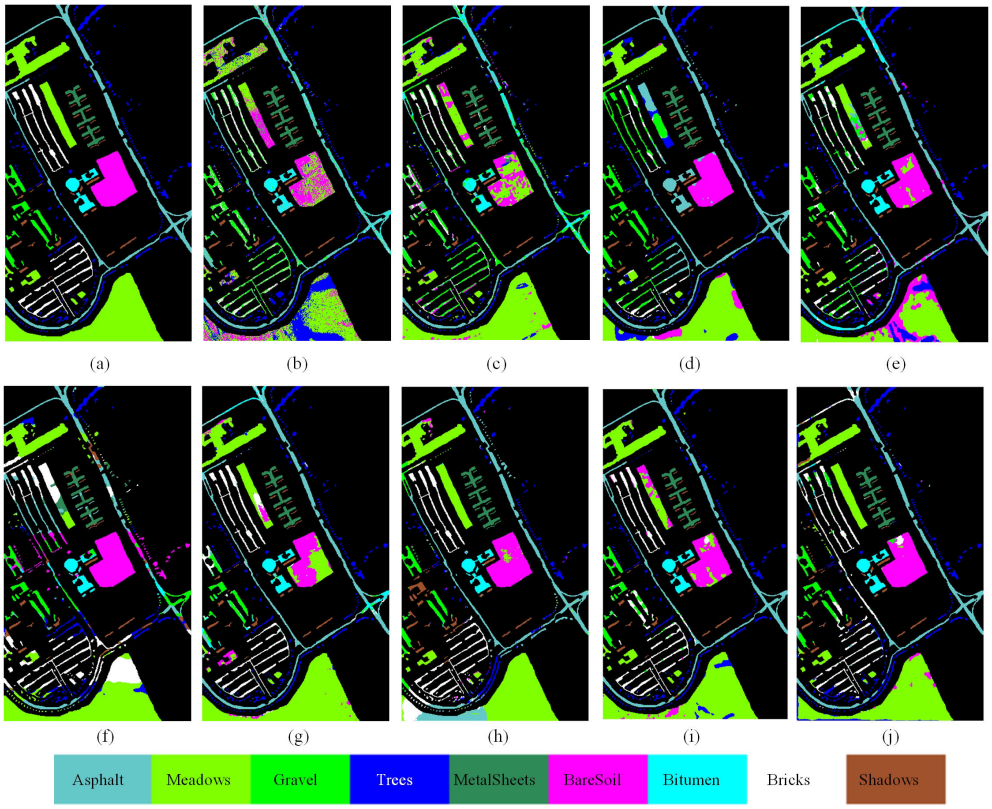


Fig. 13. Visual classification results of Pavia University from compared methods [(a) Ground truth, (b) 2D-CNN, (c) spectral former, (d) SSRN, (e) SSFTT, (f) SSGRN, (g) SC-EADNet, (h) DMVL + SVM, (i) S3Net, and (j) ours].

TABLE V
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE SALINAS DATASET (TEN LABELS PER-CLASS)

Class	2D-CNN	Spectral Former	SSRN	SSFTT	SSGRN	SC-EADNet	DMVL+SVM	S3Net	Ours
1	97.949	97.949	**100.000**	99.900	85.643	98.158	90.618	94.097	95.848
2	99.112	93.864	**100.000**	**100.000**	99.004	95.894	99.973	**100.000**	**100.000**
3	91.862	76.043	**100.000**	**100.000**	97.406	99.140	99.697	**100.000**	**100.000**
4	98.772	97.688	99.783	98.121	90.318	99.857	61.437	**100.000**	98.916
5	98.388	85.082	94.790	93.553	96.627	97.162	**99.874**	98.463	98.726
6	98.607	98.177	99.848	**99.949**	97.949	87.168	98.213	**99.949**	**99.949**
7	99.328	98.487	98.963	**100.000**	98.627	**100.000**	**100.000**	**100.000**	99.664
8	46.550	56.958	88.225	70.056	**98.233**	84.589	96.235	85.632	89.646
9	99.080	97.061	99.919	99.903	98.579	99.484	**100.000**	98.660	**100.000**
10	89.290	40.636	98.011	97.613	79.070	91.489	**100.000**	96.542	97.368
11	95.085	74.386	0.000	**99.905**	94.518	96.255	96.182	95.274	99.716
12	99.531	83.412	84.246	99.687	90.141	99.792	99.456	**100.000**	99.113
13	97.903	**100.000**	99.558	99.117	95.364	**100.000**	93.923	93.709	91.170
14	97.453	91.604	98.679	93.208	93.491	**99.720**	81.965	98.113	99.340
15	75.847	47.382	78.465	95.564	78.879	60.952	**99.843**	88.509	95.329
16	92.154	82.638	99.833	98.776	**100.000**	91.976	99.559	98.164	98.386
OA(%)	83.694	75.775	91.616	92.423	93.346	89.150	96.390	94.499	**96.529**
AA(%)	92.307	82.585	90.020	96.585	93.365	93.900	94.811	96.695	**97.698**
$\kappa \times 100$	81.991	73.135	90.644	91.602	92.584	87.910	95.988	93.883	**96.141**

to conventional contrastive learning methods. For the second ablation experiment, we eliminated the similarity-guided module to restrict the contrastive learning loss generated solely by individual sample disparities instead of those between sample groups.

Our experimental group consisted of the similarity-guided module and intergroup contrastive loss proposed in this study, which together form the complete model. Fig. 15 presents the experimental results of both ablation models and the complete model. As the number of training samples increased from 10 to 80, all three models exhibited a trend of rising OA. Nonetheless, it was consistently the case that the complete model outperformed the other two ablation models. When the number of training samples is 10, the complete model gives

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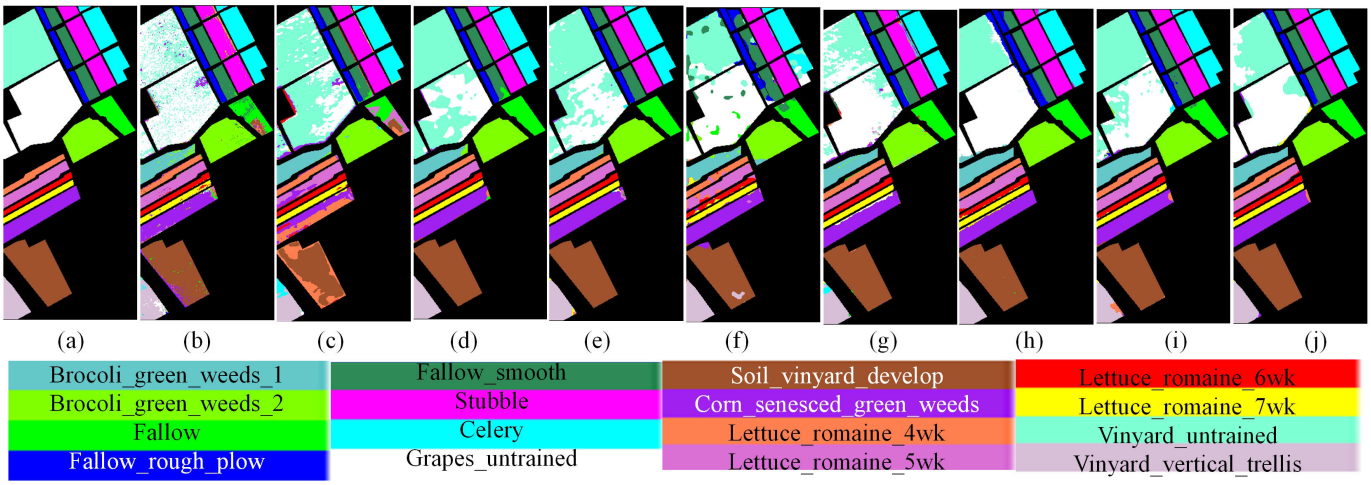


Fig. 14. Visual classification results of Salinas from compared methods [(a) Ground truth, (b) 2D-CNN, (c) spectral former, (d) SSRN, (e) SSFTT, (f) SSGRN, (g) SC-EADNet, (h) DMVL + SVM, (i) S3Net, and (j) ours].

TABLE VI
TIME CONSUMING OF DIFFERENT METHODS ON DATASETS

Dataset	Time Span (ms)								
	2D-CNN	Spectral Former	SSRN	SSFTT	SSGRN	SC-EADNet	DMVL+SVM	S3Net	Ours
IP	1465	2981	4802	3158	113	1938	3160	9807	1243
PU	4584	16334	24806	11008	499	5444	10848	84078	7378
SA	2474	14130	22280	5596	298	6726	16870	48627	7795

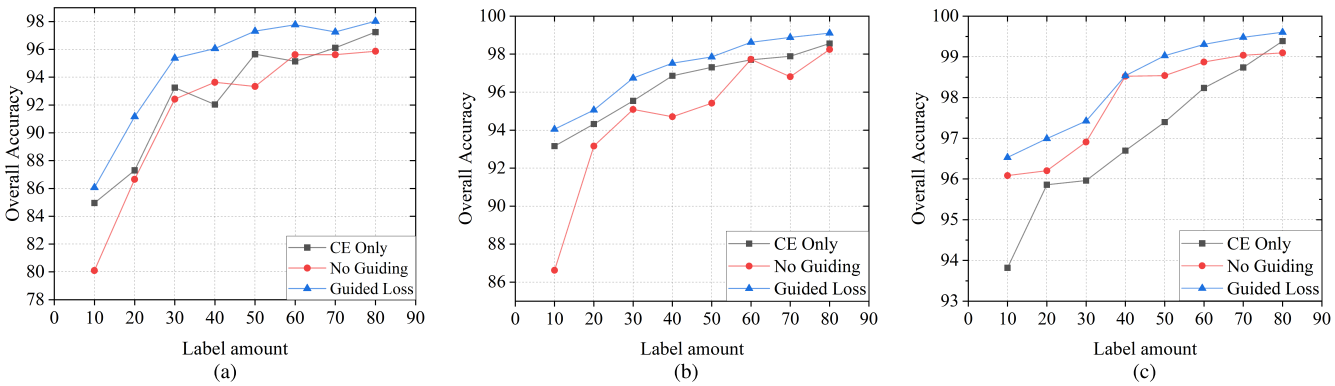


Fig. 15. Effect of guiding module on the datasets and different label amounts. (a) OAs on the Indian Pines dataset. (b) OAs on the Pavia University dataset. (c) OAs on the Salinas dataset.

TABLE VII
ABLATION RESULTS

Metrics	**Indian**			**Pavia**			**Salinas**		
	CE Loss Only	No Group Guiding	Guided Group Loss	CE Loss Only	No Group Guiding	Guided Group Loss	CE Loss Only	No Group Guiding	Guided Group Loss
OA	84.954	80.097	86.064	93.157	86.621	94.047	93.819	96.085	96.529
AA	92.221	89.765	90.297	92.108	88.600	92.576	97.338	97.704	97.698
$\kappa \times 100$	83.017	77.734	84.184	90.993	82.732	92.115	93.135	95.656	96.141

the most outstanding advances, which is shown in Table VII in detail.

Our proposed complete model on the IP dataset, which is composed of ten training samples, demonstrated an OA advantage of 1.11% and 5.967% against the pure CELoss model and the intergroup contrastive loss model without similarity guidance, respectively. This advantage was accompanied by a corresponding κ index advantage of 1.167 and 6.45. Furthermore, the pure CELoss model consistently outperformed the comparison loss model without similarity guidance in OA in

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most cases (sample sizes of 10, 20, 30, 50, 70, and 80). The lack of similarity guidance in the intergroup contrastive loss model could have hindered its ability to accurately process the features of a significant amount of unlabeled sample data, placing emphasis instead on the subtle differences between similar data rather than their shared characteristics.

On the PU dataset, composed of ten training samples, our proposed complete model had an OA advantage of 0.89% and 7.426% compared to the pure CELoss model and the intergroup comparison loss model without similarity guidance, respectively. In addition, our model had an AA advantage of 0.468% and 3.976%, and a corresponding κ index advantage of 1.122 and 9.383. As the sample size escalated from 10 to 80, the pure CELoss model continued to surpass that of the intergroup comparison loss model without similarity guidance, further highlighting the significance of intergroup guidance.

On the SA dataset, which consisted of ten training samples, our proposed complete model demonstrated an OA advantage of 2.71% and 0.444%, and a corresponding κ index advantage of 3.006 and 0.485 over the pure CELoss model and the intergroup contrastive loss model without similarity guidance, respectively. Notably, the intergroup contrastive loss model without similarity guidance excelled overall in accuracy for sample sizes ranging from 10 to 80, with particular advantage for the SA dataset, which is relatively simple with minimal differences between similar data, revealing the advantages of the contrastive loss model.

2) Feature Separability: In the proposed framework, supervised information guides unsupervised information into contrastive learning. The labels of the supervised information are used to calculate CELoss and complete back-propagation. Through this guiding process, unsupervised information can generate group similarity. To demonstrate the guiding effect of supervised signals, we recorded the feature vectors of unsupervised signals trained by CELoss and then visualized them using T-distributed stochastic neighbor embedding (TSNE) clustering. Taking the result of the IP dataset as an example, the clustering diagram of TSNE is shown in Fig. 16. It shows the grouping results of unsupervised signals guided by the supervised signals in the IP dataset. From top left to bottom right, the guiding epochs are increasing. It can be seen that at the start of the guidance, the distribution of intragroup features is discrete and that of intergroup features is mixed. After multiple rounds of guidance (i.e., 50 and 100 epochs), the intragroup features begin to gather [i.e., class 2 (red), class 10 (orange), class 11 (light blue)]. In addition, the overlap of intergroup features decreased, and the distance between groups continued to increase (i.e., classes 2 and 11 and classes 3 and 11). The above process proves the effectiveness of guiding by the supervised signals.

Compared to the classical contrastive learning methods, the GGC Loss is the key improvement of our research. The classical idea of contrastive learning is to compare the augmented samples one-on-one, which can easily generate negative pairs that belong to the same class but are treated as different classes in training. According to (4), it can be seen that the numerator part of the loss function is the similarity between samples within the group, while the denominator

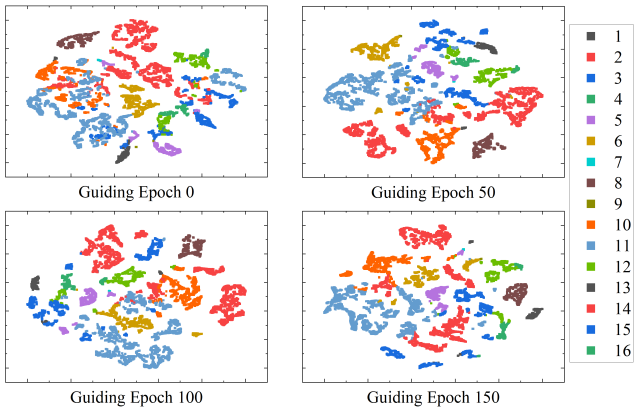


Fig. 16. Embedded feature vectors by TSNE during the guiding process.

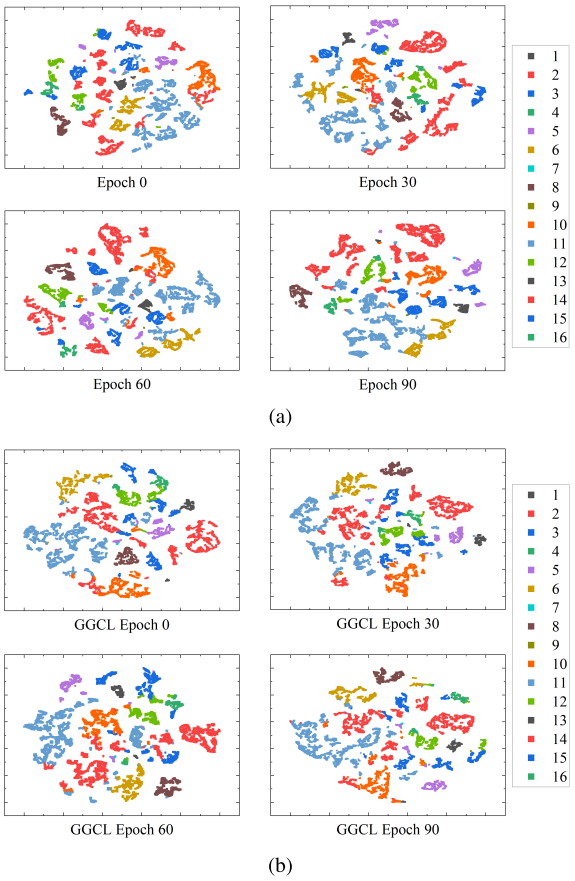


Fig. 17. Embedded feature vectors by TSNE during the training process. (a) Classical contrastive loss. (b) GGC loss.

part includes the similarity between samples between groups. During the process of loss reduction, the similarity of intragroup samples continues to increase, while the similarity of intergroup samples continues to decrease, thus achieving intragroup feature aggregation and intergroup feature separation. In order to intuitively show the improvement of GGCL, we recorded feature vectors of the unlabeled samples trained by the classical contrastive loss and GGC loss, respectively. Then, these feature vectors were visualized using the TSNE method. The following figure shows the visualization effect of TSNE dimensionality reduction using a taking the IP dataset as an example, the results are shown in Fig. 17.

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TABLE VIII
OAs OF SC-EADNET, DMVL + SVM, S3NET, AND GGCL USING DIFFERENT SAMPLE SPLITS

Models	Round	OA (%)									
		Split1	Split2	Split3	Split4	Split5	Split6	Split7	Split8	Split9	Split10
SC-EADNet	1	79.716	71.147	69.836	75.918	76.502	71.982	80.350	79.158	75.054	78.950
	2	77.343	69.379	72.685	77.076	78.492	70.576	80.929	76.941	79.050	78.680
	3	76.510	70.565	73.534	75.471	78.199	72.030	79.758	77.197	78.661	79.521
	4	77.94	71.900	72.418	73.520	80.123	71.250	81.268	75.294	77.419	77.247
	5	79.716	70.415	71.858	75.085	75.963	72.569	80.596	78.085	78.947	77.509
DMVL+SVM	1	80.213	69.821	71.938	81.364	79.627	75.032	82.428	78.495	71.441	78.964
	2	71.314	63.586	69.977	76.066	65.85	67.060	77.686	71.382	73.763	70.817
	3	77.276	67.958	68.973	78.310	69.324	69.880	73.344	72.963	64.689	74.134
	4	72.202	70.817	73.968	78.544	65.899	72.719	79.520	67.928	62.591	72.963
	5	74.632	60.221	67.694	80.017	72.787	72.729	80.993	72.231	69.490	78.193
S3Net	1	83.616	74.428	78.194	86.609	79.066	82.367	84.728	82.129	83.13	85.658
	2	85.033	73.932	77.173	80.394	81.604	82.456	85.224	82.793	82.803	85.925
	3	82.654	76.281	76.013	82.179	78.323	83.507	86.750	84.587	82.605	85.568
	4	86.024	74.606	79.354	84.27	77.530	82.724	85.264	83.973	82.803	85.430
	5	84.815	76.232	78.333	82.317	79.354	82.327	86.858	83.259	83.328	85.410
GGCL (ours)	1	84.022	**86.302**	82.079	85.856	**83.279**	81.485	86.094	85.747	83.913	87.095
	2	86.520	83.417	82.555	**87.610**	83.180	82.962	86.322	82.654	**85.023**	87.571
	3	**89.434**	83.180	81.217	85.618	82.862	83.279	86.322	81.891	83.695	**88.552**
	4	84.389	85.132	**84.508**	83.269	82.545	**84.409**	87.729	**86.064**	82.872	84.954
	5	86.867	85.658	79.076	84.934	83.170	82.575	**88.165**	84.518	81.881	84.538

From the comparison of the two subfigures in Fig. 17, it can be observed that after the same training epochs, clusters in Fig. 17(a) are increasingly concentrated, which is a positive signal. However, in Fig. 17(a), even if the clusters belong to the same class, their distances continue to increase so they distribute increasingly dispersed. In Fig. 17(b), although each cluster is relatively loose compared with Fig. 17(a), clusters belonging to the same class are moving closer to each other. This indicates that group guidance is beneficial for the aggregation of the same class, rather than just the aggregation of the same cluster. The result demonstrates the effectiveness of the GGC loss used in this article.

3) Impact of Dataset Split on Accuracy: In our experiments, each class provided ten labeled samples for supervised training and provided the rest unlabeled samples for unsupervised training. The selection of labeled samples is random so that the model may be significantly influenced by the different supervised information. In order to analyze the impact caused by the selection of labeled samples on the model performance, we randomly split the whole dataset ten times and obtained different labeled training sets. Each labeled set consists of ten labeled samples from each class. Then, the different splits of datasets were input into the SC-EADNet, DMVL + SVM model, S3Net, and our GGCL model. Taking the Indian Pines dataset as an example, the OA is shown in Table VIII.

From Table VIII, it can be seen that in all ten random splits, our GGCL model achieved the maximum OA values compared to the other three supervised contrastive learning models. Furthermore, we calculated the average OA values of Table VIII, which as shown in Fig. 18. From Fig. 18, it can be seen that on all ten split training sets, the average OA of our GGCL model is the highest among that of the other three

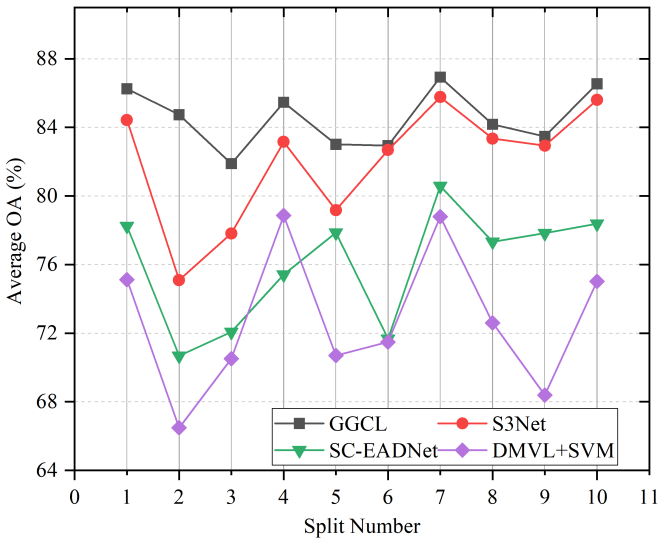


Fig. 18. Round-average OAs of SC-EADNet, DMVL + SVM, S3Net, and GGCL using different sample splits.

models. The average OA of models varies on different training sets, and they exhibit similar fluctuation patterns, which can be seen as the influence caused by few-shot label selection. For example, on the 7th and 10th splits, all models have higher OA values than their performance on the 3rd and 5th splits. In spite of that, on all dataset splits our GGCL model is superior to other models, verifying its independent accuracy advantages on the dataset.

V. CONCLUSION

This article proposes an end-to-end framework that can adaptively integrate unsupervised information into supervised contrastive learning. The proposed GGCL uses a

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similarity-guided module to measure the spectral–spatial similarity of unsupervised samples based on supervised signals and group them according to this similarity. The key feature of this method is to end-to-end integrate unsupervised information with supervised signals. As the training epoch increases, the unsupervised information gradually integrates with the grouping information under the guidance of the supervised signal, ultimately improving the classification performance.

In the future, we will investigate frameworks that combine supervised and unsupervised contrastive learning in more realistic environments where hyperspectral data may contain label noise, class imbalance, and domain shift. Developing robust and efficient contrastive learning algorithms that can address these challenges is crucial for practical HSI classification applications.

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